

中文论文大纲 (AI4Math)

标题 (建议)

从自然语言到可验证推理: 基于语义规约的AI数学可靠性框架

1. 引言 (Introduction)

- 背景: 大语言模型 (如 GPT-4、Claude) 在数学推理中的能力与局限
- 问题: 自然语言推理难以直接验证 → “靠谱性”问题
- 现状:
 - 纯生成 (LLM reasoning)
 - 事后验证 (proof checking, symbolic tools)
- 核心论点:

可靠性不能仅依赖“最终验证”，必须引入中间语义规约层

2. 问题分层模型 (Three-Layer Decomposition)

提出三层结构:

2.1 第一层: 生成 (Generation Layer)

- LLM生成自然语言推理链 (CoT)
- 特点: 概率性、非类型化、非严格语义

2.2 第二层: 语义规约 (Semantic Reduction Layer)

- 将自然语言映射为结构化表示:
 - 逻辑表达式 (FOL / HOL)

- 程序 (Python / DSL)
- 定理证明对象 (Proof Objects)
- 类型化中间表示 (Typed IR)
- 对应理论基础:
 - Curry-Howard correspondence
- 核心挑战:
 - 语义不稳定 (ambiguity)
 - 类型不完整 (missing typing)
 - 映射不可逆 (non-invertibility)

2.3 第三层: 验证 (Verification Layer)

- 在形式系统中验证:
 - 自治性 (consistency)
 - 可推导性 (derivability)
 - 事实一致性 (factual grounding)
 - 约束满足 (constraint satisfaction)
- 工具:
 - Coq
 - Agda
 - SMT / symbolic solvers

3. 核心问题: 语义规约瓶颈 (The Bottleneck)

- 观察：

当前系统失败主要源于第二层，而非生成或验证
 - 分析：
 - 验证依赖“可计算对象”
 - 自然语言 → 形式对象转换不稳定
 - 结果：
 - 验证失效 (no handle to verify)
 - 幻觉无法检测
-

4. 方法框架(Proposed Framework)

4.1 结构化中间表示(Structured IR)

设计统一Typed IR, 包括：

- 类型系统 (Type System)
- 操作语义 (Operational Semantics)
- 约束系统 (Constraints)
- 可组合结构 (Composable Graph / AST)

候选形式：

- DSL (领域特定语言)
- AST (抽象语法树)
- Proof Objects
- Workflow Graph

4.2 生成 → IR → 验证 → 渲染 Pipeline

提出标准流程：

Natural Language (LLM)



Typed IR (semantic reduction)



Formal Verification



Natural Language Rendering

关键思想：

- 自然语言只作为 interface
- 内核为可验证结构

4.3 双向一致性 (Bidirectional Consistency)

- IR ↔ Natural Language
- 要求：
 - 可解释性 (interpretability)
 - 可回溯性 (traceability)

5. 与现有工作的关系 (Related Work)

- LLM + Tool (Toolformer, Code Interpreter)
- Program-of-Thought / Chain-of-Thought
- Proof assistants (Coq, Lean)
- Neuro-symbolic systems

差异：

- 本文强调 语义规约层作为核心系统组件
-

6. 在AI4Math中的应用 (Applications)

6.1 自动定理证明 (ATP)

- LLM生成 proof sketch $\rightarrow$ IR $\rightarrow$ Coq验证

6.2 数学解题系统

- Word problem $\rightarrow$ symbolic program $\rightarrow$ solver

6.3 教学与评估 (AI+Education)

- 学生解答 $\rightarrow$ IR $\rightarrow$ 自动评分
-

7. 实验设计 (Experimental Setup)

- 数据集：
 - GSM8K / MATH
 - 对比：
 - 纯LLM vs IR-based system
 - 指标：
 - 正确率 (accuracy)
 - 可验证率 (verifiability)
 - 一致性 (consistency)
-

8. 讨论(Discussion)

- 局限：
 - IR设计复杂
 - 语义映射成本高
- 扩展：
 - 通用推理系统
 - Agent系统 (NEOLAF/KSTAR架构)

9. 结论(Conclusion)

- 核心结论：

可靠AI推理 = 生成 + 语义规约 + 验证
- 关键贡献：
 - 明确提出“语义规约层”为核心瓶颈
 - 提供结构化IR路径

English Paper Outline (AI4Math)

Title (Suggested)

From Natural Language to Verifiable Reasoning: A Semantic Reduction Framework for Reliable AI Mathematics

1. Introduction

- Background: LLMs (e.g., GPT-4, Claude) in mathematical reasoning
- Problem: Natural language reasoning is not directly verifiable
- Current approaches:
 - Generation (LLMs)
 - Post-hoc verification
- Thesis:

Reliability cannot rely solely on final verification; a semantic reduction layer is required

2. Three-Layer Model

2.1 Generation Layer

- Produces natural language reasoning (CoT)
- Probabilistic, untyped, informal

2.2 Semantic Reduction Layer

- Translates natural language into:
 - Logic
 - Programs
 - Typed IR
 - Proof objects
- Theoretical grounding:
 - Curry-Howard correspondence
- Challenges:

- Ambiguity
- Missing typing
- Non-invertibility

2.3 Verification Layer

- Checks:
 - Consistency
 - Derivability
 - Factual correctness
 - Constraint satisfaction
- Tools:
 - Coq
 - Agda

3. The Bottleneck: Semantic Reduction

- Main failure source lies in Layer 2
- Without stable reduction:
 - Verification has no anchor
 - Hallucinations persist

4. Proposed Framework

4.1 Typed Intermediate Representation (IR)

- Components:
 - Type system
 - Operational semantics
 - Constraints
 - Compositional structure

Forms:

- DSL
 - AST
 - Proof objects
 - Workflow graphs
-

4.2 Pipeline Architecture

Natural Language



Typed IR



Verification



Natural Language Rendering

Key idea:

- Natural language = interface
 - IR = execution core
-

4.3 Bidirectional Alignment

- IR ↔ Language

- Ensures:
 - Interpretability
 - Traceability
-

5. Related Work

- Chain-of-Thought, Program-of-Thought
- Tool-augmented LLMs
- Proof assistants
- Neuro-symbolic AI

Contribution:

- Elevates semantic reduction as the central layer
-

6. Applications

6.1 Automated Theorem Proving

- LLM $\rightarrow$ proof sketch $\rightarrow$ IR $\rightarrow$ verification

6.2 Math Problem Solving

- Word problems $\rightarrow$ symbolic programs

6.3 AI Education

- Student answers $\rightarrow$ IR $\rightarrow$ grading
-

7. Experiments

- Benchmarks: GSM8K, MATH
 - Metrics:
 - Accuracy
 - Verifiability
 - Consistency
-

8. Discussion

- Limitations:
 - IR design complexity
 - Mapping difficulty
 - Extensions:
 - General reasoning systems
 - Agent architectures
-

9. Conclusion

- Core claim:

Reliable AI reasoning requires a structured semantic layer
- Contribution:
 - Formalizes the 3-layer architecture
 - Identifies semantic reduction as the key bottleneck

